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Experimental study on the magnetic permeability of inclusion filled soft polymeric composite for soft-core transformer applications

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ABSTRACT

The magnetic permeability of three kinds of inclusions (Carbonyl iron, Ferroferric oxide, NiFe-alloy) filled polymer (Polydimethylsiloxane) is investigated at room temperature by using a Buck circuit for the dynamic test and vibrating sample magnetometer (VSM) for the static test. Unlike the traditional measurement method using an impedance analyzer, the self-built circuit can generate a maximum test current of ampere amplitude over high frequencies to simulate the transformer's operation status. The present work considers the effects of inclusion content (from 0 VOL.% to 33 VOL.%) and testing frequencies(100 kHz-1MHz). The experimental results of permeability are compared with the theoretical predictions in the previous works. Three main conclusions can be drawn: First, the magnetic permeability increases with the inclusion addition; Second, as the testing frequency increases, the permeability of the composite material maintains constant, showing no evident dependence on the testing frequency; Third, type of magnetic inclusion will gently affect the permeability of the composite material. This work systematically investigated the permeability of polymer-based magnetic materials, and the results are helpful for the soft-core transformer application of magnetoactive elastomers.

1. Introduction

Magnetoactive elastomers are composed of distributed magnetic inclusions and the surrounding soft elastic matrix [1] [–] [3]. Generally, the magnetic components are constituted of metallic, ferrite, and alloy particles alone or mixed, whereas the polymer matrix is inherently non-magnetic [4] and non-conductive [5]. Compared with the sintered magnets, magnets made of magnetoactive elastomers have many advantages, such as outstanding performance in high frequency alternating electromagnetic fields [6], or they can sustain the shape integrity without being broken after an intense force of impact [7]. Therefore, many researchers are concerned about using magnetoactive elastomers to replace sintered cores of transformers as their engineering applications [8,9].

Magnetic permeability is an essential parameter of magnetoactive elastomers for applications [10]. In general, the impedance analysis method is the most commonly employed for measuring the permeability of SMCs [11] [–] [14]. The technique lies in recording ring-shaped samples' impedance under an AC magnetic field [6]. The impedance of a ferromagnetic core-filled coil can be calculated by considering it as an equivalent series circuit of inductors and resistors. For clarity, representative works based on this method are briefly introduced here: In 1990, Irvine et al. investigated the inductance (which is proportional to the permeability) of ZnMn ferrite toroid sample over the frequency from 5Hz to 13 MHz, using a signal of amplitude 0.1V [15]. In 2000, Carara et al. studied the permeability of the soft magnetic materials ($FeSi_{3\%}$), with the test frequency ranges from 100Hz to 100kHz and the probe current ranges from 0.1mA to 50mA. The previous experimental results indicate that a larger current amplitude may induce an increase in permeability [11]. On the one hand, the measuring voltage and current based on the impedance analysis method are usually very low at high frequencies, which are totally different from the working voltage or current of the transformers. On the other hand, there are a few numbers of experiential works on the magnetic permeability of magnetoactive elastomers [10]. Consequently, there is an urgent demand to develop a test method for measuring the permeability of magnetoactive elastomers at large currents and varied frequencies within the operating range as transformers.

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Table 1
Composition of the samples (volume percentage).

Inclusion volume fraction		0%	10%	20%	33%
Add one of the three inclusions	NiFe-alloy (g)	0	20	40	80
	Carbonyl iron (g)	0	18	38	76
	Fe_3O_4 (g)	0	12	26	None
PDMS and curing agent (g)		16			
Silicon oil (g)		4			

The test circuit in the present work is based on the Buck circuit which provides several amperes current and high-frequency magnetic field due to the high-performance switch device Gallium Nitride (GaN) transistor. Three kinds of potential soft magnetic powders (carbonyl iron, ferroferric oxide, NiFe-alloy) are mixed into Polydimethylsiloxane (PDMS) matrix at different volume fractions (from 10% to 33%), and the composite is expected to exhibit large and stable permeability in the testing frequencies from 100 kHz to 1MHz. The imaginary part of the permeability is not considered because the testing frequencies are below MHz level, and the samples are inductive (eddy effect is neglected). For comparison, the static magnetic characteristics at room temperature are measured by the vibrating sample magnetometer (VSM). Model constructed in COMSOL MULTIPHISICS has been used to calculate the inductance of the cylindrical core inductor, and the permeability of the sample can be therefore determined. The permeability of toroidal samples is calculated from the experiment directly without using finite element analysis (FEA) to verify the obtained results of the cylindrical samples.

The purpose of this work is to explore the feasibility of soft-core transformers by measuring the permeability of different magnetic composite materials at static and varied frequencies to guide researchers in the engineering application of magnetoactive elastomers.

2. Materials

PDMS and curing agent SYLGARD 184 were purchased from Dow Corning (U.S.A.). Pt -catalyst (68,478-92-2) for the curing of PDMS and silicon oil PMX-200 (10 mPa · s) were purchased from Aladdin Co. Ltd. (China). Carbonyl iron powders (CIPs) were bought from Xindun Alloy Co Ltd. (China). NiFe-alloy (Fe70Ni30) powders were purchased from Yuhang Metal Material Co. Ltd. (China). Fe_3O_4 powders were bought from Yuexin Chemical & Plastic Firm (China).

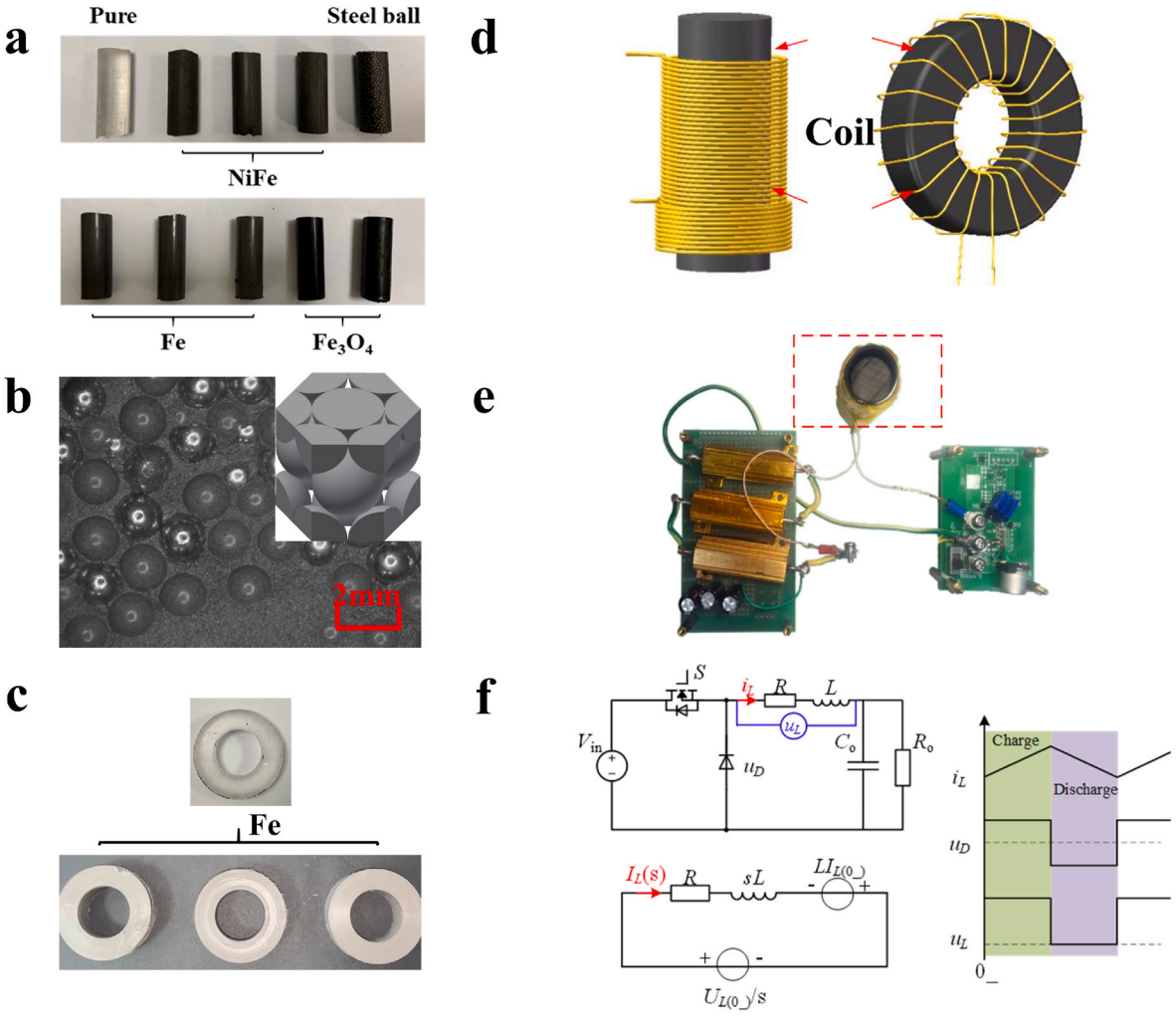


Fig. 1. Test samples and circuit. (a) Cylindrical samples: In the image above, from left to right are pure PDMS, samples with 10% NiFe, 20% NiFe, 33% NiFe, and 20% NiFe sample with steel balls; In the image below, from left to the right, are samples with 10% CIP, 20% CIP, 33% CIP, 10% Fe_3O_4, 20% Fe_3O_4. The radius of cylindrical pieces is 9.8 mm, and the height is 53 mm. (b) Image of steel ball sample captured by electronic magnifier (WST-3800, China). Scale bar, 2 mm. The image in the right corner is the 3D schematic diagram of the close-packed hexagonal structure. (c) Toroidal samples: pure PDMS, samples with 10% CIP, 20% CIP, 33% CIP. The outer diameter is 57.7 mm, and the inner diameter is 31.8 mm. The height of the sample is 15 mm. (d) Wires around the magnetic core. (e) Test circuit board. The image in the red box is the test sample. (f). Buck circuit and circuit model in the complex frequency domain. The variation of current i_L, voltage u_{Diode} andu_L are depicted in the dotted line diagram. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
The Electrical parameters of the buck circuit.

Parameter	Specification
Input voltage	30V
Switching frequency	100 kHz-1MHz
Duty ratio	0.5
Load resistance	5Ω
Switching device	TI LMG5200 (GaN)

Fabrication of the magnetic sample: 14.5g PDMS and 15 μL Pt -catalyst were firstly mixed by planetary mixer (ARE-310, U.S.A.), then 1.5g curing agent SYLGARD, 4g silicon oil, and magnetic powders of a certain amount (Table 1) were immersed into the mixture, followed by mixing for the second time (1500 rpm, 1 min). Subsequently, the mixture was placed into a vacuum to remove air bubbles. The obtained precursor was poured into the mold and cured for 4 h at 65 °C in the oven.

Fabrication of dense-packed sample: Steel balls (diameter 1.5 mm) were put into the mold, and we vibrated the mold by the vibrating machine (VORTEX-Kylin-Bell 5) to ensure particle dense-packed. Then magnetic precursor (20% NiFe-alloy) was poured into the mold to fill the air gap, and then vacuum was applied to remove air bubbles. This step was repeated several times until the mold was filled. Finally, the mixture was cured for 4 h at 65 °C in the oven. For the mold used in the experiment, the weight concentration of different components is given: steel balls 79 wt% and magnetic precursor 21 wt%.

3. Test methods and principles

3.1. Test processes

The pure PDMS sample is fabricated as the control group. Samples filled with NiFe-alloy, NiFe-alloy with steel balls, CIP and Fe_3O_4 are prepared as the experimental groups, respectively. As shown in Fig. 1a, the microstructure of the steel ball group is observed under an optical microscope. The diameter of the steel balls is 1.5 mm, and the arrangement is similar to the close-packed hexagonal structure (Fig. 1b). Samples fabricated in the cylindrical form will facilitate the experimental operation (for the convenience of wrapping wires) but also lead to inevitable problems. The magnetic lines of induction pass through two different materials (the magnetic core and the outside air) to form a closed magnetic circuit. Then flux leakage appears due to the same order of permeability magnitude shared by the two materials. These facts disturb the proportional relationship between inductance and permeability, affecting the accuracy of the experiment. Samples of the toroidal structure are considered to obtain a more accurate experimental result (Fig. 1c). Both test samples are elastic and deformable, which cannot be broken after impact. All the samples are wrapped with wires (diameter 0.6 mm). The total turns of cylindrical and toroidal pieces are 50 and 80, respectively (Fig. 1d).

Table 2 shows the electrical parameters of the Buck circuit, which steps down its input DC voltage into a lower output DC voltage by turning the switching device on and off periodically. The Buck circuit has been applied extensively in the field of high current applications. Besides, the magneto-electrical apparatus such as inductor and transformer operates at hundreds of kHz up to several MHz [16]. So in the experiment, the testing frequency is set from 100 kHz to 1 MHz. Fig. 1e demonstrates the experimental circuit in the present work, and the equivalent circuit for the inductor in the complex frequency domain is shown in Fig. 1f.

3.2. Experiment measuring the inductance

The coil can be regarded as an electrical element that connects an ideal inductor L and a resistor R in series. For linear time-invariant inductance components, the terminal voltage is the product of the inductance value$L$ and the derivative of the current to time$\frac{di_L}{dt}$. The corresponding circuit equations are generally differential equations. The time-invariant linear circuits are usually transformed from the time domain to the complex frequency domain. As shown in the top image of Fig. 1f, the terminal voltage of the coil is:

$$u_L = i_L R + L\frac{di_L}{dt} \tag{1}$$

In Eq. (1), u_L is the constant input voltage. The circuit under the complex frequency domain is shown in the bottom image of Fig. 1f, and the corresponding equation is:

$$(sL+R)I_L - Li_L(0_-) = \frac{u_L(0_-)}{s} \tag{2}$$

Here, s is the complex frequency, $u_L(0_-)/s$ is the voltage u_L after Laplace transformation and current I_L in the complex frequency domain can be obtained by Eq. (3):

$$I_L = \frac{u_L(0_-)}{R}\left(\frac{1}{s} - \frac{1}{s+\frac{R}{L}}\right) + \frac{i_L(0_-)}{s+\frac{R}{L}} \tag{3}$$

The current i_L in the time domain can be obtained by performing inverse Laplace transformation on Eq. (3):

$$i_L(t) = \mathscr{L}^{-1}(I_L(s)) = \frac{u_L(0_-)}{R} + \left[i_L(0_-) - \frac{u_L(0_-)}{R}\right]e^{-\frac{R}{L}t} \tag{4}$$

Hence the amplitude of current ripple can be obtained:

$$\Delta i_L(t) = i_L\left(\frac{T}{2}\right) - i_L(0) = \left[i_L(0_-) - \frac{u_L(0_-)}{R}\right]\left[e^{-\frac{R}{L}\frac{1}{2f}} - 1\right] \tag{5}$$

The current ripple amplitude is equal during the charging and discharging process, and the sign is opposite. When the duty ratio is 50%, Eq. (5) can be transformed into:

$$i_L(0_-)_{Char} - \frac{u_L(0_-)_{Char}}{R} = -\left[i_L(0_-)_{Dischar} - \frac{u_L(0_-)_{Dischar}}{R}\right] \tag{6}$$

In Eq. (6), $i_L(0_-)_{Char}$ is the charging current and$i_L(0_-)_{Dischar}$ is the discharging current. $u_L(0_-)_{Char}$ is the voltage corresponding to the peak charging current, and $u_L(0_-)_{Dischar}$ is the opposite. Therefore, the resistance of the coil can be obtained:

$$R = \frac{u_L(0_-)_{Char} + u_L(0_-)_{Dischar}}{i_L(0_-)_{Char} + i_L(0_-)_{Dischar}} \tag{7}$$

Inductance L can be obtained from Eq. (5):

$$L = -\frac{R}{2f \cdot In\left(\frac{i_L(0_-)_{Dischar} - i_L(0_-)_{Char}}{i_L(0_-)_{Char} - \frac{u(0_-)_{Char}}{R}} + 1\right)} \tag{8}$$

Eq. (8) is more suitable for situations where the internal resistance of the inductor cannot be ignored. For example, the eddy current of the steel ball sample increases obviously with the increase of frequency.

According to the volt-second balance principle, the voltage amplitude across the inductance is equal, and the sign is opposite. (During the charging and discharging process, with a duty ratio of 50%). The inductance L can be obtained since Δt and Δi_L is also known from the experiment:

$$L = u_L^{'}\frac{\Delta t}{\Delta i_L} \tag{9}$$

$u_L^{'}$ in Eq. (9) is the induced voltage generated by the inductance according to Lenz's law. Here, the voltage $u_L^{'}$ can be obtained:

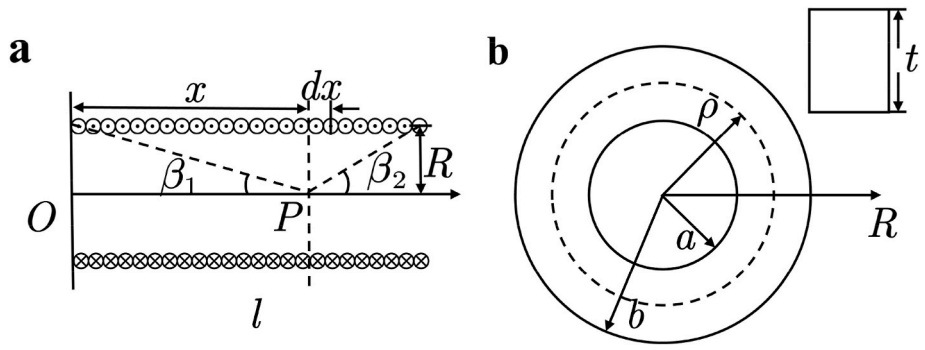


Fig. 2. Sample geometry schematic. (a) Section view of the cylindrical sample. (b) Toroidal sample and its section view.

$$u_L^{'}=\frac{1}{2}(u_{LChar}-u_{LDischar}) \tag{10}$$

In Eq. (10), u_{LChar} and $u_{LDischar}$ can be obtained by the oscilloscope. In the subsequent sections, the inductance values of the steel ball group are calculated based on Eq. (8), and the inductance value of the remaining groups are calculated based on Eq. (9), considering the influence of the skin effect.

3.3. Calculation of magnetic permeability

For the cylindrical sample (Fig. 2a), the wire is evenly and tightly wound on its surface. From Biot-Savart Law, the magnetic flux density in the hollow coil along its axis can be obtained by integral as

$$B=\frac{\mu_0 NI}{2l}\left\{\frac{x}{(x^2+R^2)^{\frac{1}{2}}}+\frac{l-x}{((l-x)^2+R^2)^{\frac{1}{2}}}\right\} \tag{11}$$

Here μ_0 is the magnetic permittivity in the vacuum, N is the total turns of wires, I is the current flowing across the coil. l is the height of the hollow coil and R is the radius. The number of flux linkage passing through the coil section area S is:

$$\Psi=\frac{\mu_0 N^2 IS}{l^2}\left[(l^2+R^2)^{\frac{1}{2}}-R\right] \tag{12}$$

The inductance of the coil is:

$$L=\frac{\Psi}{I}=\frac{\mu_0 N^2 S}{l}\left[\left(1+\left(\frac{R}{l}\right)^2\right)^{\frac{1}{2}}-\frac{R}{l}\right] \tag{13}$$

When there is a magnetic core inside the solenoid, the inductance can be approximately expressed as:

$$L=\frac{\Psi}{I}=\frac{\mu_0 \mu_r N^2 S}{l}\left[\left(1+\left(\frac{R}{l}\right)^2\right)^{\frac{1}{2}}-\frac{R}{l}\right] \tag{14}$$

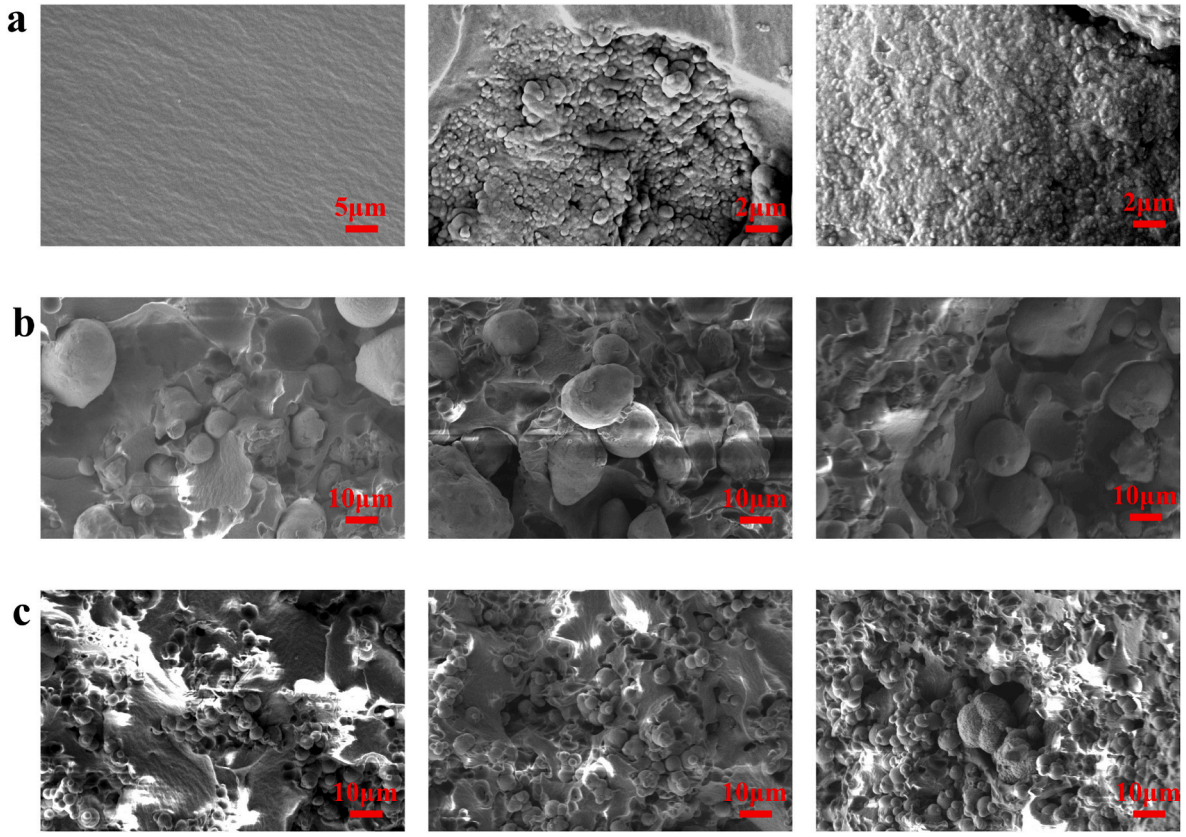


Fig. 3. SEM images of different samples. (a) Pure PDMS, samples with 10% Fe_3O_4 and 20% Fe_3O_4. (b) Samples with 10% NiFe, 20% NiFe, 33% NiFe. (c) Samples with 10% CIP, 20% CIP, 33% CIP.

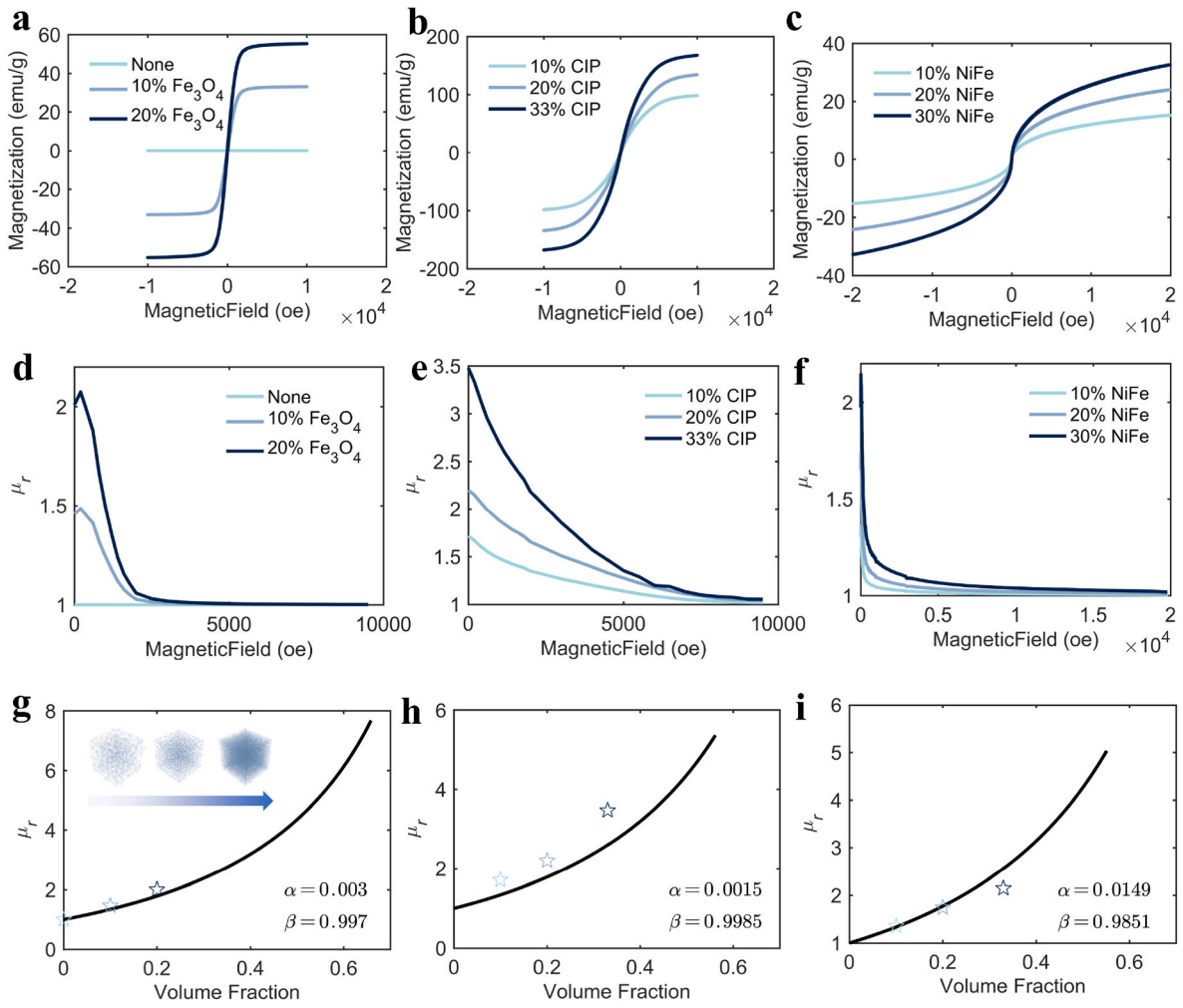


Fig. 4. VSM results and theoretical permeability. (a)–(c): M − H curves of the samples obtained by VSM. (d)–(f): The magnetic permeability versus applied magnetic field obtained by VSM. (g)–(i): Comparison between theoretical magnetic permeability and initial magnetic permeability obtained by VSM data (Colorized Stars).

As mentioned above, Eq. (14) is suitable for the case of high permeability magnetic core ($\mu_r \gg 1$). However, when the permeability of the material and air are in the same order, it may not be applicable.

As shown in Fig. 2b, according to the Ampere circuital theorem, the magnetic field distribution inside the magnetic ring along the radial direction is not uniform.

$$2\pi\rho H_\rho = NI \tag{15}$$

The number of flux linkage passing through the coil is calculated as:

$$\Psi = N\int_S \vec{B}\,d\vec{S} = \int_0^t \int_a^b \frac{N^2 I\mu_0\mu_r}{2\pi\rho}\,dz\,d\rho = \frac{\mu_0\mu_r N^2 It}{2\pi} In\frac{b}{a} \tag{16}$$

The inductance can be approximately expressed as:

$$L = \frac{\Psi}{I} = \frac{\mu_0\mu_r N^2 t}{2\pi} In\frac{b}{a} \tag{17}$$

By measuring the inductance of the magnetic core coil and the air core coil, their ratio is the relative permeability.

4. Results and discussion

4.1. Microstructure of samples

Fig. 3 is the microstructure of the samples captured by SEM. From Fig. 3a, the tear section of the pure piece is very smooth, and no extra impurities can be observed. With the increasing addition of magnetic inclusions, particle aggregation is getting more notable. The diameter of Fe_3O_4 powders is around 200 nm, and the size distribution is even. In Fig. 3b, it can be found that the size of NiFe-alloy particles ranges from 5 Ψ to 50μm. Fig. 3c shows that the average diameter of CIP is about 5 μm. The maximum volume fraction of the magnetic inclusions is 33% for the CIP and NiFe samples, and the polymer matrix can be regarded as an adhesive to bond isolated particles together. Fe_3O_4 particles without surface modification cannot be cured into the polymer matrix. In the experiment, we used oleic acid modified powders as inclusions. However, the sample with 33% Fe_3O_4 cannot be fabricated due to the inadequate precursor curing.

4.2. Static magnetic properties of the samples

The static magnetic properties of the test samples were measured using VSM. Fig. 4a, b, and 4c show the initial magnetization curves of Fe_3O_4, CIP, and NiFe groups. In the first group, the sample begins to saturate when the external field reaches 1600oe. For two different inclusion contents, the saturation magnetization is 33 emu/g and 55 emu/g, respectively. The saturation magnetization of pure Fe_3O_4 particles is 65 emu/g [17], larger than that of the composite material. In the second group, with the increased applied magnetic field, the curve does not show a clear starting point of saturation. The saturation magnetization is 98 emu/g, 134 emu/g, 167 emu/g for the samples of three contents, respectively. In the third group, it can be obtained that the applied

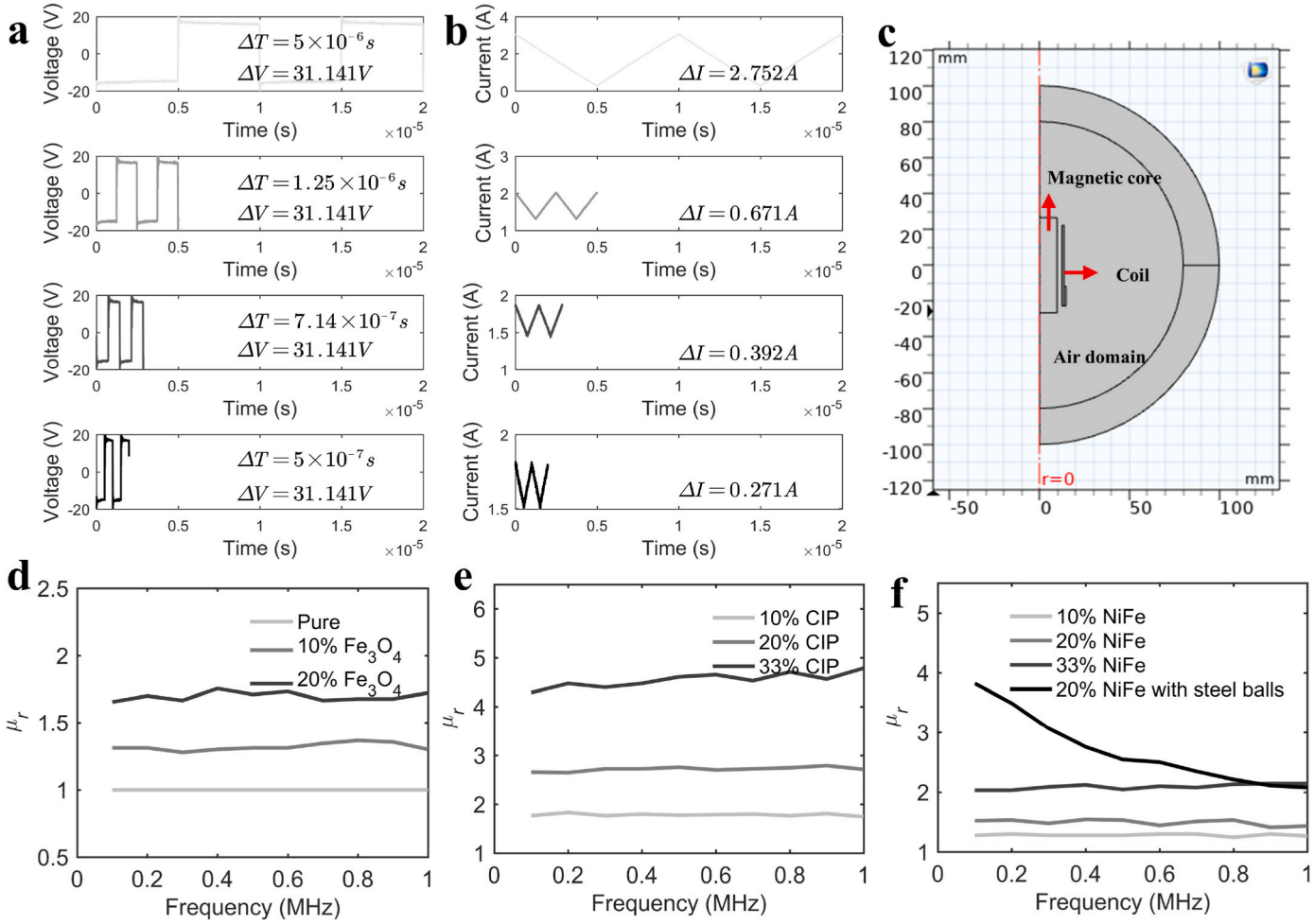


Fig. 5. Test parameters of the Buck circuit and simulated permeability of cylindrical samples. (a) The voltage applied on the coil (Pure sample) and the test frequencies from top to bottom are 0.1 MHz, 0.4 MHz, 0.7 MHz, 1 MHz. (b) Current flowing out of the coil (Pure sample) (c). Model constructed in COMSOL. (d) The simulated magnetic permeability of the pure sample and samples with Fe_3O_4 particles. (e) Samples with Fe particles. (f) Samples with NiFe-alloy particles and steel balls.

magnetic field to saturate the material should be greater than 20,000 oe. By finding the slope of the initial magnetization curve, the relationship between the relative permeability of the materials and the applied magnetic field can be obtained. In Fig. 4d, the initial magnetic permeability varies from 1.456 to 2.009 for the samples with different Fe_3O_4 content. In Fig. 4e, the initial magnetic permeability is 1.716 for the sample with the lowest CIP content, 2.2 for the higher, and 3.477 for the highest. The magnetic permeability decreases monotonically with the applied magnetic field. Compared with the first group, the magnetic permeability of the sample decreases slowly with the applied magnetic field, indicating that the samples are difficult to reach saturation. Fig. 4f shows that the magnetic permeability of NiFe samples decreases drastically with the external magnetic field. The initial magnetic permeability is 1.357, 1.740, 2.145 for the samples with different NiFe contents, respectively. By the way, samples of the same inclusion but different volume fractions exhibit the same intrinsic coercivity H_{cj}: 50oe for Fe_3O_4 particles filled samples, 8oe for CIP filled samples, and 12oe for NiFe particles filled samples.

The theoretical magnetic permeability of the composite can be expressed as [18]:

$$\mu_r = \frac{\varphi\mu_1\alpha\left(1+\frac{\varphi}{4}\beta^2\right)+(1-\varphi)\mu_0}{\varphi\alpha\left(1+\frac{\varphi}{4}\beta^2\right)+(1-\varphi)} \tag{18}$$

In Eq. (18), φ and μ_1 is the volume fraction and magnetic permeability of the inclusion, α andβ are two dimensionless parameters:$\alpha = 3\mu_0/(\mu_1+2\mu_0)$, $\beta = (\mu_1-\mu_0)/(\mu_1+2\mu_0)$. As shown in Fig. 4g, the theoretical permeability fit well with the experimental results for the Fe_3O_4 group. However, for samples with CIPs or NiFe alloy inclusions, the theoretical permeability is lower than the experimental one for the former and larger for the latter.

4.3. Experimental results of cylindrical samples

Square wave with the same amplitude and different frequencies was applied on the empty coil, and the inductor current was recorded (Fig. 5a). It is observed from Fig. 5b that the current ripple decreases with the increase of frequency, and the coil inductance can be calculated based on Eq. (9) with the differences of the voltage and current during the charging and discharging processes. The model of the coil is constructed in COMSOL (Fig. 5c). The simulated inductance can be obtained by the sweep simulation of changing the core magnetic permeability (1 for the empty coil). When the simulated inductance is the same as the measured inductance, the corresponding magnetic core permeability is regarded as the permeability in the experiment. The model geometry is verified by calculating the example with PDMS core. When the simulated inductance is the same as the experimental inductance (PDMS permeability is 1), the model geometry is adopted in the subsequent simulation where the core is magnetic. It can be obtained in Fig. 5d that the sample permeability with 10% Fe_3O_4 particles and 20% Fe_3O_4

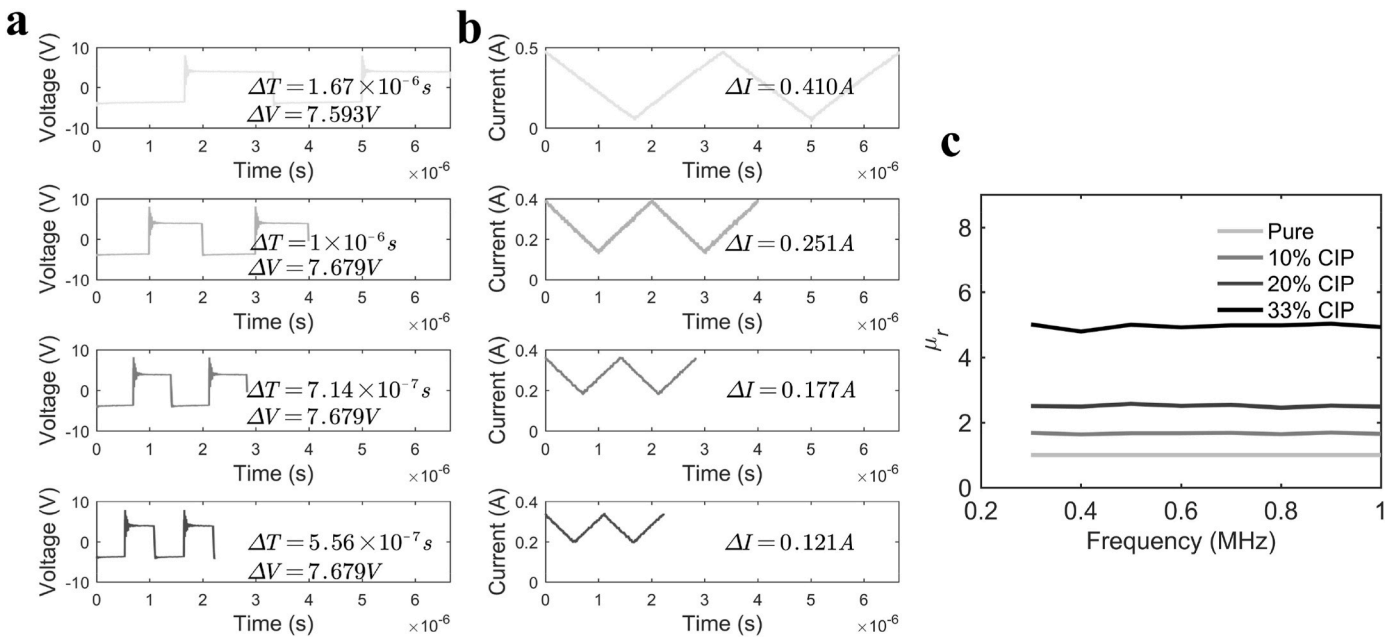


Fig. 6. Test parameters of the Buck circuit and simulated permeability of toroidal samples. (a) The voltage applied on the coil (Pure sample) and the test frequencies from top to bottom are 0.3 MHz, 0.5 MHz, 0.7 MHz, 0.9 MHz. (b) Current flowing out of the coil (Pure sample). (c) Simulated permeability of samples with CIP particles.

Table 3
Summary and comparison of the similar works of transformer application.

Reference	Materials	Frequency range	Magnetic permeability (Real part)	Mechanical characteristics
[19]	Mn–Zn ferrite	1 MHz	3000	Hard
[20]	Silicon steel	100Hz-1MHz	500–10	Hard
[21]	Electrical steel	0–1 MHz	870–35	Hard
This work	Carbonyl–iron powder, PDMS	0.1 MHz–1MHz	Around 4.9	Soft

particles is 1.3 and 1.7, which doesn't change with the frequency. Fig. 5e shows the magnetic permeability of the sample with 10% CIP particles, 20% CIP particles, and 33% CIP particles is 1.8, 2.7, and 4.6, respectively. Fig. 5f shows the magnetic permeability of the sample with 10% NiFe particles, 20% NiFe particles, and 33% NiFe particles is 1.3, 1.5, and 2.1, respectively. It is worth noting that the permeability of the sample with steel balls decreases from 3.8 to 2.1 with the frequency, which indicates that as the frequency increasing, eddy currents appear inside the steel balls. The purpose of adding steel balls is to increase the volume fraction of the magnetic particles. However, the particle size of the magnetic conductive particles will also affect the magnetic permeability of the composite material under high frequency. Therefore, to avoid the generation of eddy currents in practice, particles with high permeability and high resistivity should be used, or conductive particles with small sizes (micron or nanometer) should be used. Compared with the static result, the permeability of the Fe_3O_4 and NiFe samples is identical. The simulated permeability of the piece with a high Fe volume concentration is larger. Toroidal specimens are used in the experiment to obtain a more accurate measurement result.

4.4. Experimental results of toroidal samples

Similar to the test method mentioned in section 4.3, the voltage applied to the coil and the current flowing out are shown in Fig. 6a and b. From Eq. (9), the inductance of the hollow coil with the pure PDMS core can be obtained as approximately 15 uH. The magnetic coil inductance can also be calculated. The magnetic permeability of the composite can be obtained by calculating the ratio of and based on Eq. (17). For the sample with 10% CIP, 20% CIP, 33% CIP, the permeability is 1.7, 2.5, and 4.9, respectively. There is a neglectable difference between the cylindrical result and the toroidal result. Compared with the static results obtained from VSM, the obtained magnetic permeability for the sample with 33% inclusion is higher. Up to now, we cannot explain it. We summarize the results of previous related works as transformer applications, as shown in Table 3. Compared with the traditional materials, the magnetoactive elastomers exhibit much less magnetic permeability, but they are more stable over varied frequencies. Hence, for transformers that work at different frequencies, cores made of magnetoactive elastomers are reasonable.

5. Conclusions

This work investigates the permeability of magnetoactive elastomers as soft-core transformer applications. On the theoretical basis of the impedance analysis, the circuit we developed can measure the permeability of both cylindrical (by combining finite element simulation) and toroid samples under different frequencies (100 kHz-1MHz). All the inclusion-filled samples show neglectable permeability decreasing as the testing frequency increases, indicating that the permeability is frequency insensitive (only valid for the frequency range in this work) and eddy current is avoided, except for the steel balls filled one. Sample filled with CIPs exhibits the maximum permeability (4.9 for 33 vol%), and sample filled with ferroferric oxide is the easiest to be saturated. In contrast, the opposite conclusion can be drawn for the sample with NiFe-alloy particles. Consequently, the magnetoactive elastomers composed of CIPs and PDMS might be reasonable for soft-core transformer applications.

Author statement

Xiaocheng Hu and Zipeng Liang contributed equally to this work. Qiyang Li and Sideng Hu guided the entire work, Shaoxing Qu participated in the manuscript edition.

Detailed contribution:

Xiaocheng Hu: Conceptualization, Methodology, Validation, Investigation, Writing.

Zipeng Liang: Conceptualization, Software, Validation, Investigation.

Qiyang Li: Review & Editing, Supervision.

Sideng Hu: Supervision.

Shaoxing Qu: Review & Editing, Supervision.

Declaration of competing interest

There are no conflicts to declare.

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